

Can Market Signals Improve Delta Hedging?

A leakage-controlled replication experiment on historical SPY paths

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We test hedge inputs – not historical option execution

**Can point-in-time
signals reduce
terminal replication
error versus
fixed-volatility
Black–Scholes?**

Controlled comparison

Same payoff, initial premium, hedge calendar, and transaction-cost rule.

OBSERVED MARKET DATA

- ▶ SPY adjusted prices and volume
- ▶ VIX, XLK returns, and short rates

MODEL-DEFINED CLAIMS

- ▶ European calls with controlled maturity and moneyness
- ▶ No historical option bids, asks, or contract IV

Four strategies; one common delta engine

FV-BS

**Fixed 21-day
realized volatility**

Benchmark fixed at
contract initiation.

ROLLING-BS

**Updated 21-day
realized volatility**

Uses the latest price
information.

VIX-BS

**Daily VIX as
hedge volatility**

Adds
forward-looking
option-market
information.

MSA-DELTA

**Rolling vol $\times$ risk
multiplier**

Combines seven
market signals.

Common premium · daily rebalancing · identical payoff · identical costs

Definition: volatility changes delta away from the money

COMMON HEDGE FORMULA

$$\Delta_t = N(d_{1,t}), \quad d_{1,t} = \frac{\ln(S_t/K) + (r + \frac{1}{2}\sigma_t^2)\tau}{\sigma_t\sqrt{\tau}}.$$

Example contract

$S = 100$, $K = 105$, $\tau = 21/252$, and $r = 3\%$.

Why this matters: volatility adjustments materially change OTM and ITM deltas, but have little effect on an ATM delta near 0.5.

NORMAL STATE

$$\sigma = 20\% \Rightarrow \Delta = 0.220$$

↓ +0.077 shares
per option

STRESS STATE

$$\sigma = 28\% \Rightarrow \Delta = 0.297$$

Definition: seven trailing signals create three risk states

STANDARDIZED RISK SCORE

$$R_t = 0.30 \text{IVRank}_t + 0.20 \text{VIX}_t + 0.20 \frac{\text{RV}_{21,t}}{\text{RV}_{63,t}} \\ + 0.10 \text{VolumeShock}_t + 0.10 \text{Drawdown}_{63,t} \\ + 0.05 \text{SectorDiv}_t + 0.05 |\Delta \text{Rate}_{21,t}|.$$

Features are standardized using development-period statistics only.

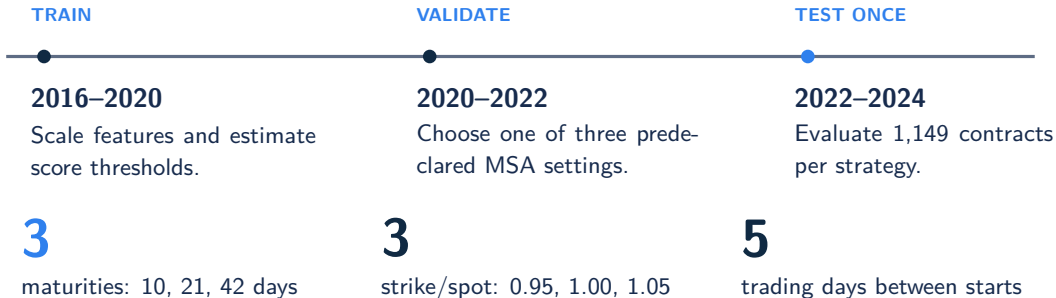
$$\text{NORMAL} \quad R_t < q_{70} \quad m = 1.00$$

$$\text{ELEVATED} \quad q_{70} \leq R_t < q_{90} \\ m = 1.15$$

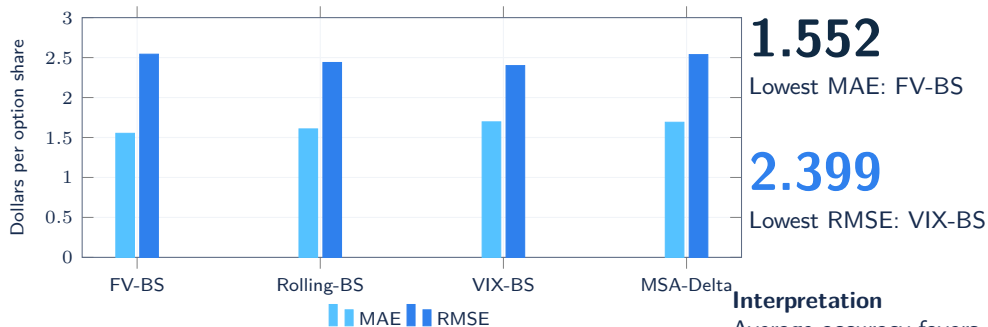
$$\text{STRESS} \quad R_t \geq q_{90} \quad m = 1.40$$

$$\sigma_t^{\text{MSA}} = m(Z_t) \hat{\sigma}_{21,t}$$

The test period stays untouched until final evaluation



No single hedge wins every error metric



Interpretation

Average accuracy favors simplicity; large-error control favors forward-looking volatility.

VIX helps most where losses are largest

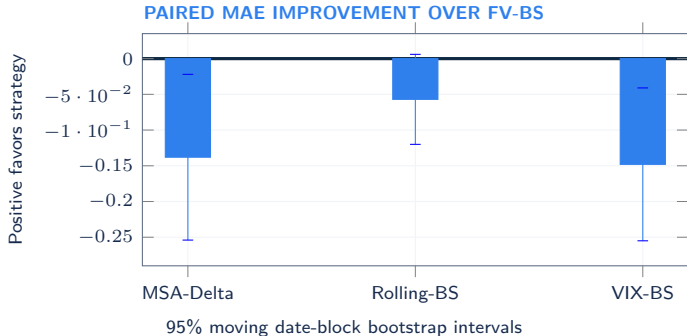
5.52

VIX-BS 95% loss Expected
Shortfall

6.75

FV-BS 95% loss Expected
Shortfall

Lower is better. VIX reduces expected loss in the worst 5% of outcomes by about **18%**.



The ranking survives transaction-cost changes

Transaction cost	Lowest MAE		Lowest RMSE	
0 bps	FV-BS	1.456	VIX-BS	2.394
5 bps	FV-BS	1.552	VIX-BS	2.399
10 bps	FV-BS	1.745	VIX-BS	2.510

STABLE FINDING 1

Fixed volatility remains best for average absolute error.

STABLE FINDING 2

VIX remains best for squared error at every cost level.

Results are also reported for all nine maturity–moneyness cells.

The useful signal is simpler than the model we designed

Forward-looking volatility improves tail-risk control – but more signals do not guarantee a better hedge.

KEEP

VIX-BS as the strongest tail-risk benchmark.

REJECT

A blanket claim that additional complexity improves hedging.

NEXT

Repeat the design using contract-level option quotes when available.

IMPORTANT LIMITATIONS

- ▶ VIX is an index-level 30-day proxy, not contract-level SPY IV.
- ▶ Controlled European claims omit historical spreads and early exercise.
- ▶ Block bootstrap mitigates, but does not eliminate, overlap dependence.